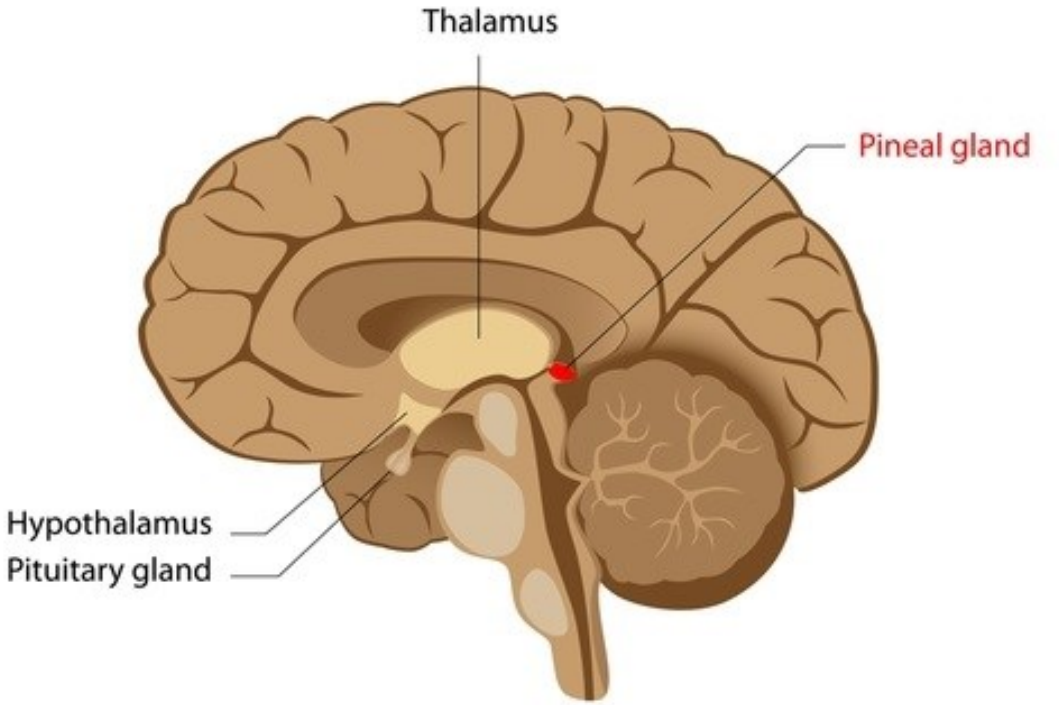
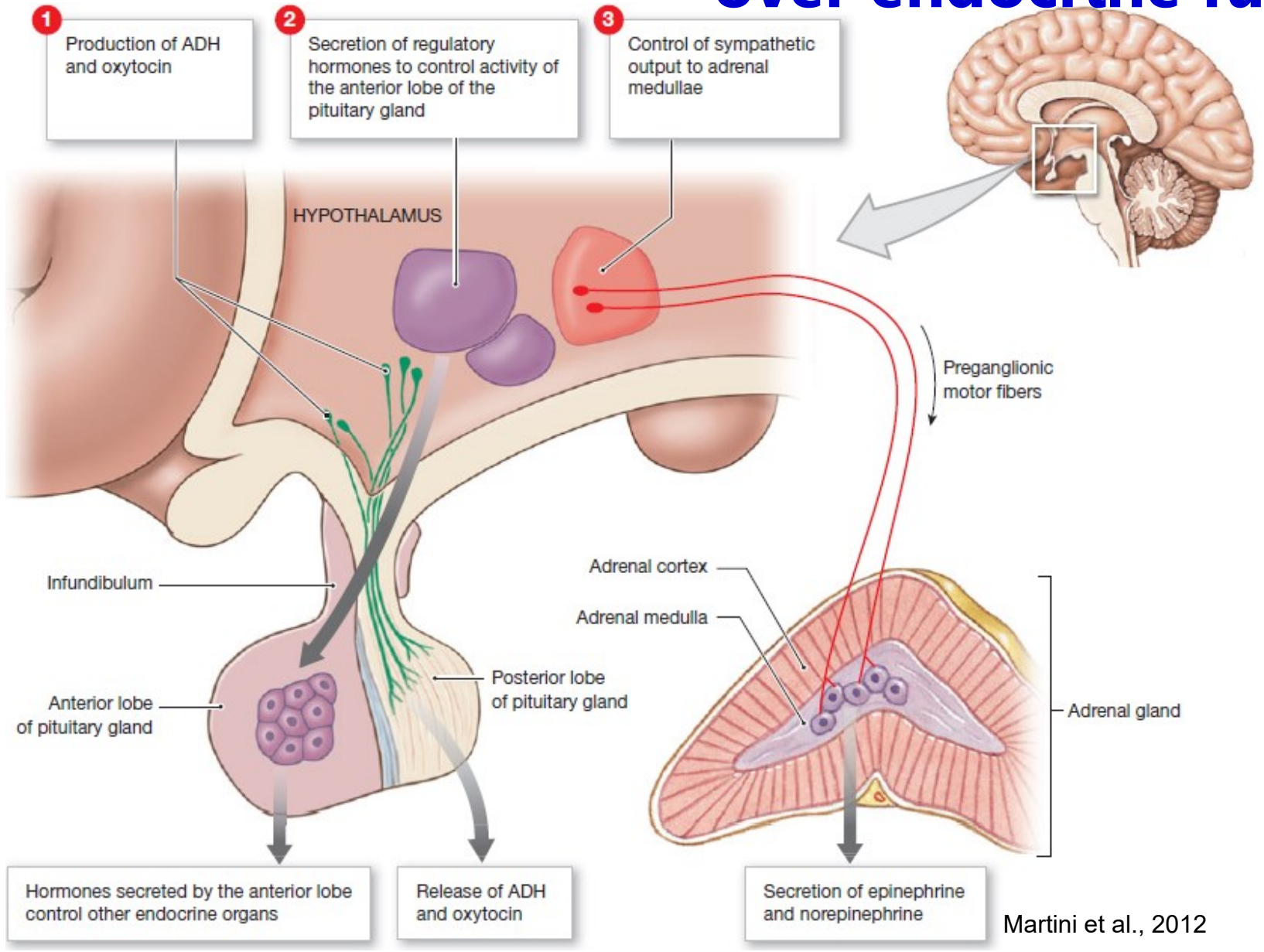
Physiology of hypothalamus, pituitary and thyroid



Assoc. Prof. Pissared Muangnil, Ph.D.
Faculty of Science, PSU

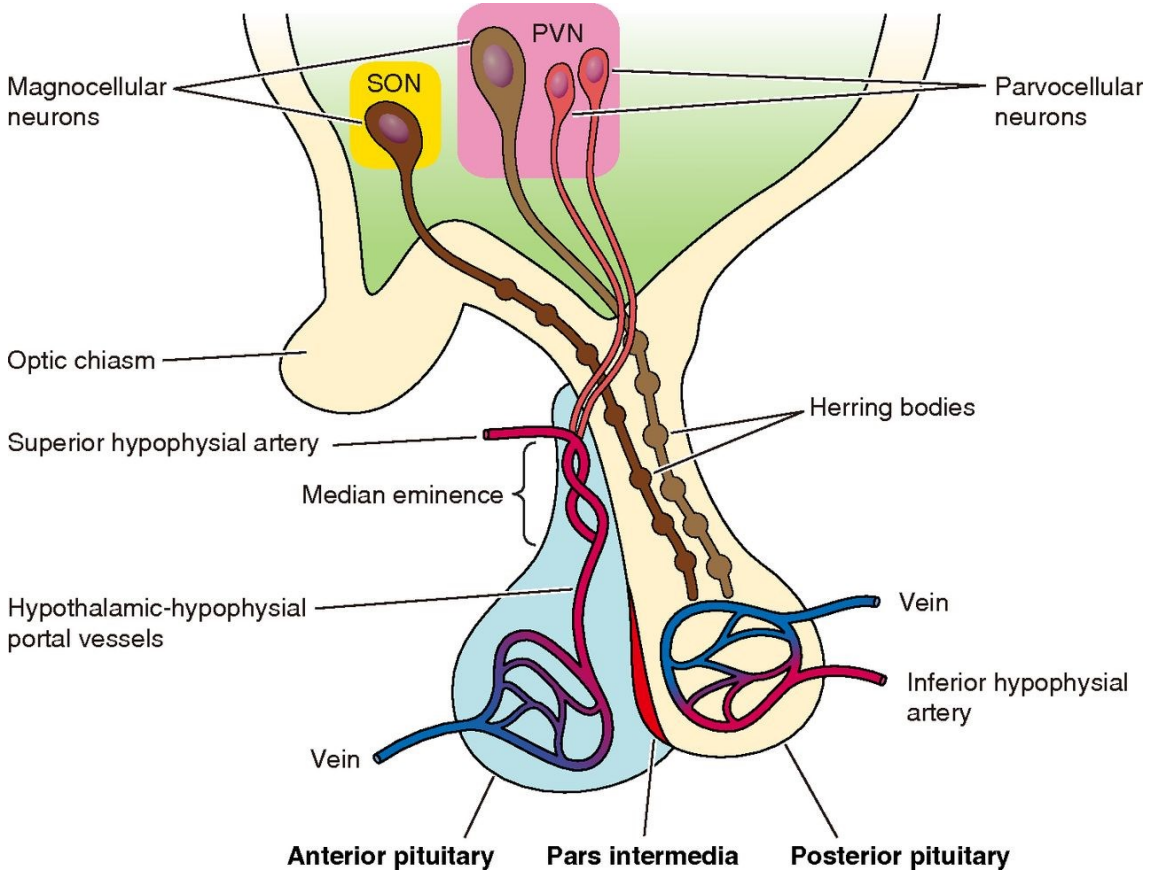
The mechanisms of hypothalamic control over endocrine function



Martini et al., 2012

2 types of neurons in the hypothalamus mediate the endocrine functions:

- **Magnocellular neurons**: located in **paraventricular & supraoptic nuclei**, produce neurohormones (**oxytocin & ADH**)
- **Parvocellular neurons**: release **releasing or inhibiting neurohormones**



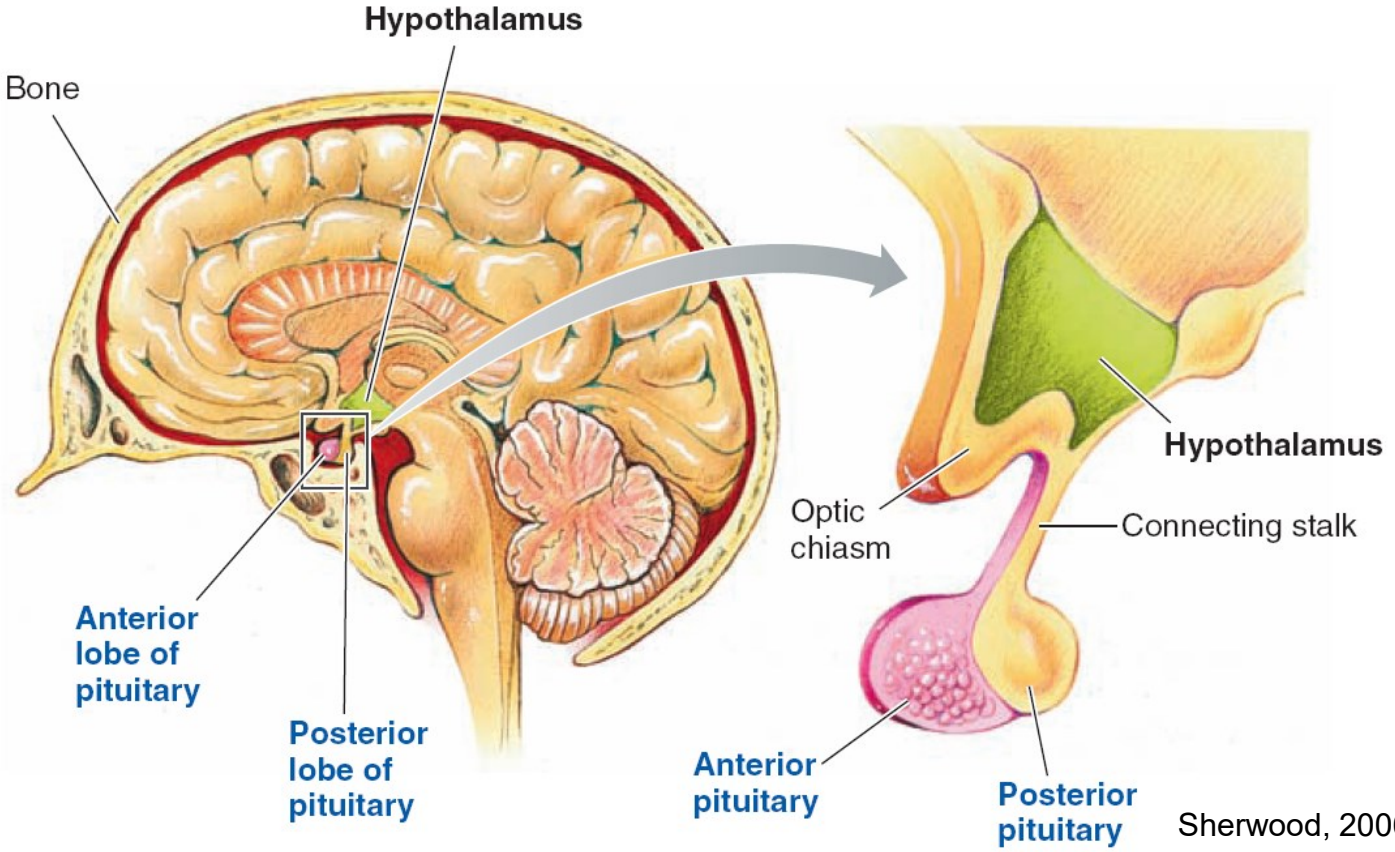
Major hypophysiotropic hormones

Hormone	Effect on the Anterior Pituitary
Thyrotropin-Releasing Hormone (TRH)	Stimulates release of TSH (thyrotropin) and prolactin (PRL)
Corticotropin-Releasing Hormone (CRH)	Stimulates release of ACTH (corticotropin)
Gonadotropin-Releasing Hormone (GnRH)	Stimulates release of FSH and LH (gonadotropins)
Growth Hormone–Releasing Hormone (GHRH)	Stimulates release of growth hormone (GH)
Growth Hormone–Inhibiting Hormone (GHIH, Somatostatin)	Inhibits release of GH and TSH
Prolactin-Releasing Hormone (PRH)	Stimulates release of PRL
Prolactin-Inhibiting Hormone (PIH, Dopamine)	Inhibits release of PRL

Sherwood, 2006

Pituitary gland

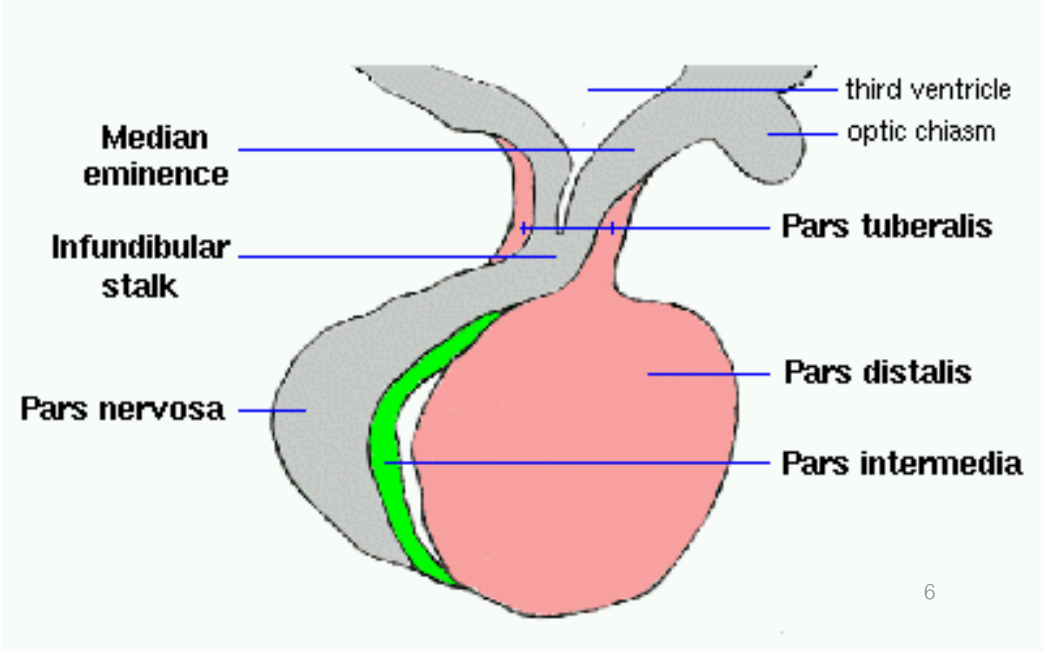
- **Hypophysis** is attached to the hypothalamus by the **pituitary stalk/hypophysial stalk** (infundibulum + Pars tuberalis).
- **Anterior lobe (adenohypophysis) & Posterior lobe (neurohypophysis)**
- 9 peptide hormones: anterior (7) & posterior (2)
- All hormones bind to membrane receptors, use cAMP as a 2nd messenger.



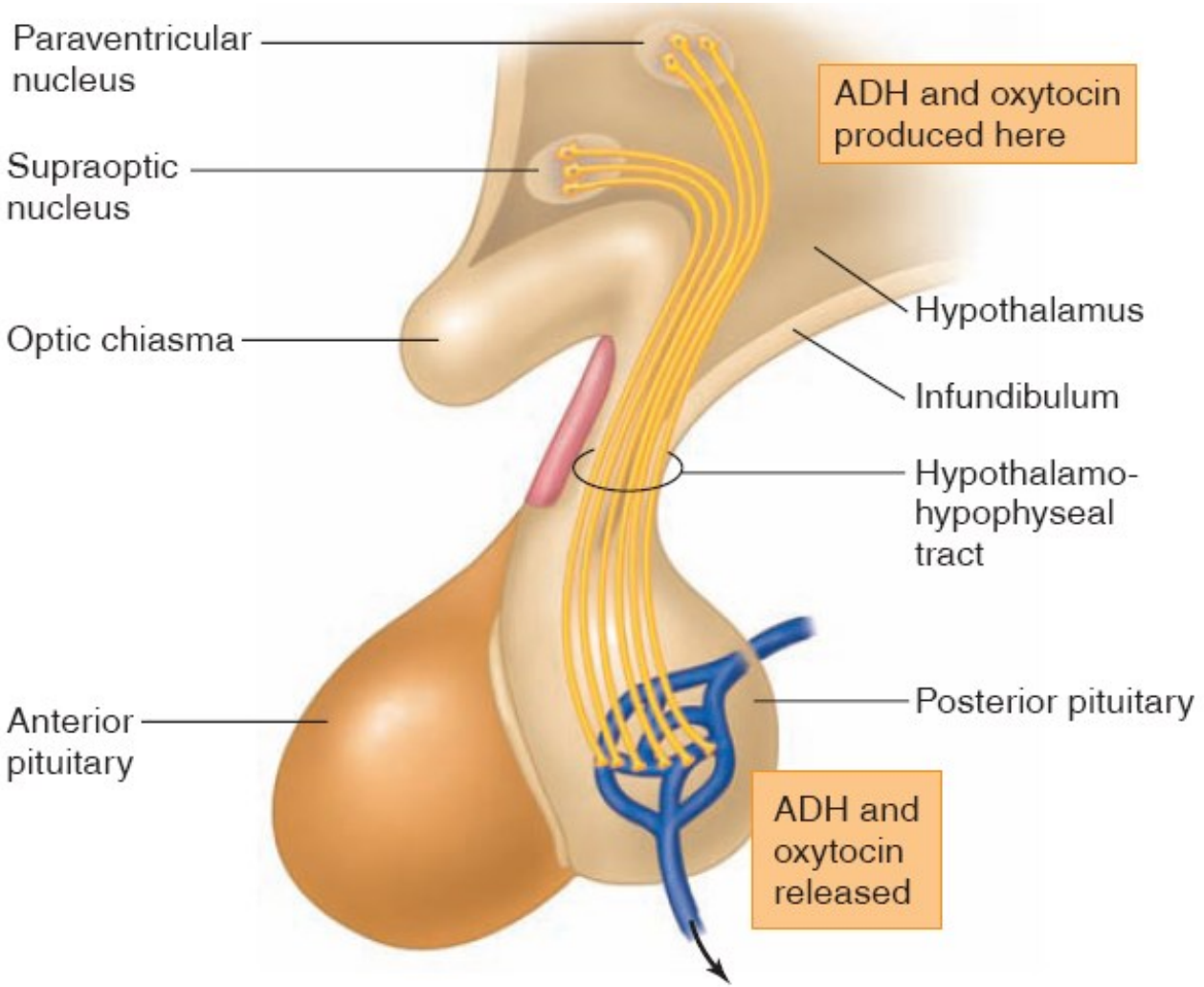
Sherwood, 2006

Posterior pituitary gland

- **Neurohypophysis:** an extension of the hypothalamus
- Consists of the **pars nervosa**
- Contact with adenohypophysis & infundibulum
- Nerve fibers extend through the infundibulum along with small neuroglia-like cells (pituicytes).
- Contains the axons terminal of **magnocellular neurons** (generate & propagate AP, and exocytosis of the contents of their secretory granules)

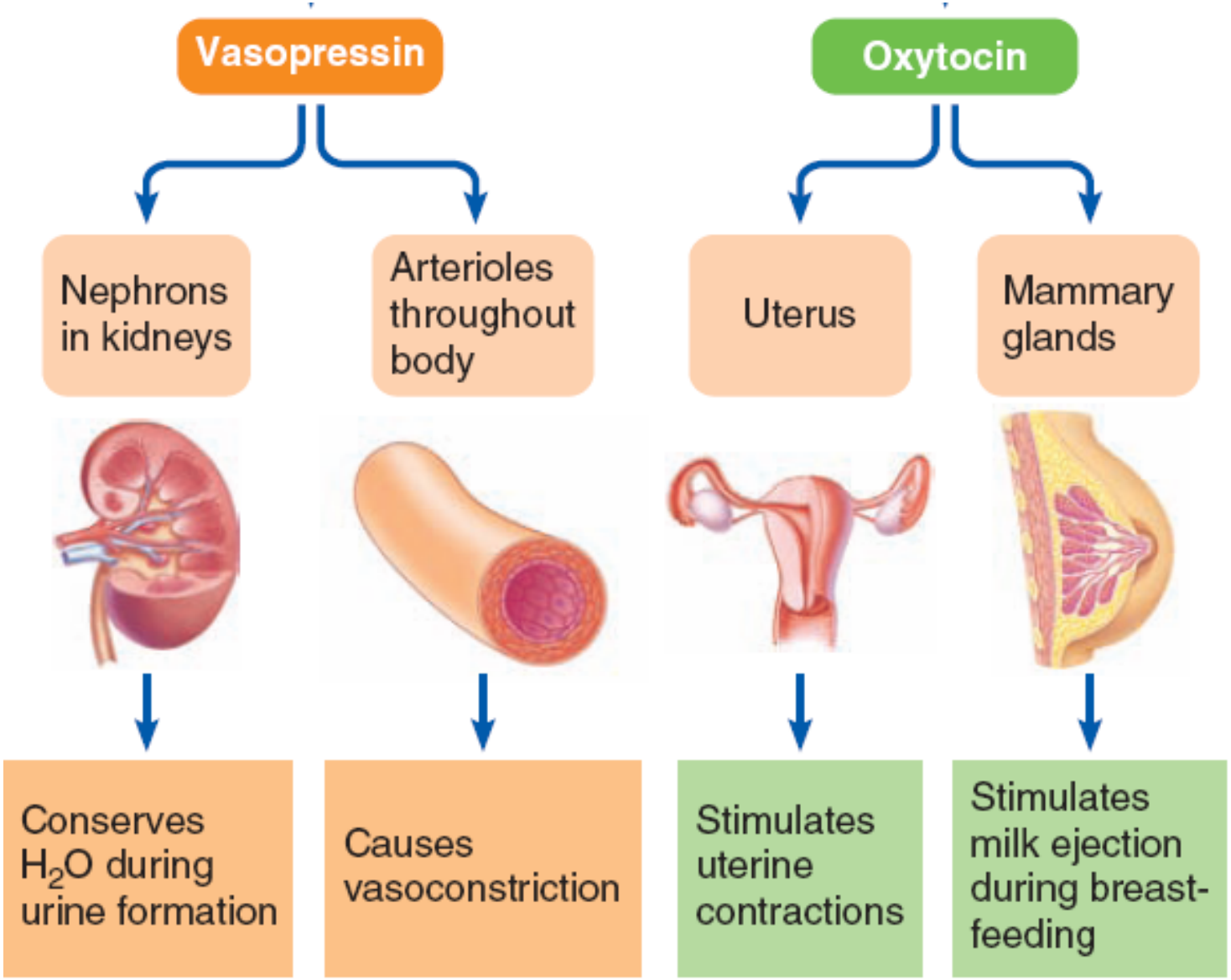


- **Stores and releases hormones** (**vasopressin & oxytocin**)
- These hormones are transported to the posterior pituitary by axons in the **hypothalamo-hypophyseal tract**.



Fox, 2011

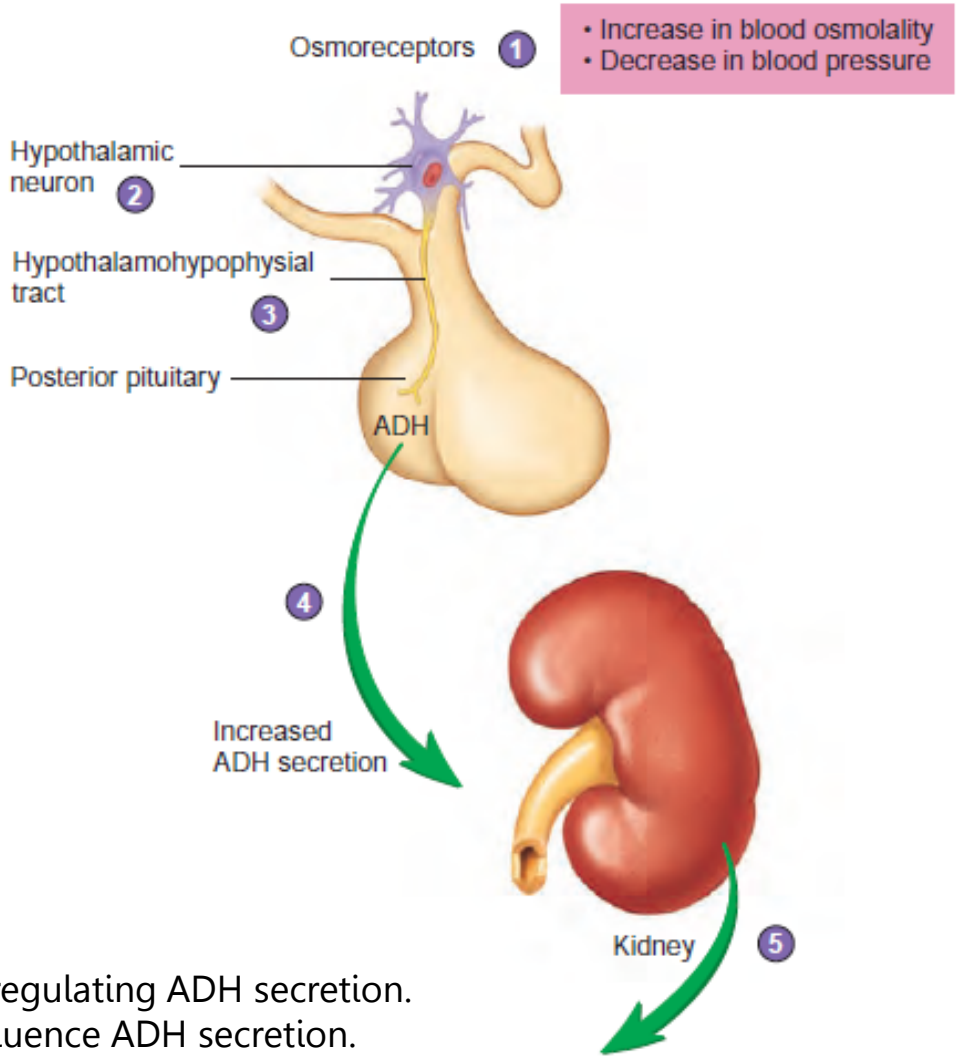
Effects of vasopressin and oxytocin



Sherwood, 2006

Control of ADH secretion

1. Osmoreceptors in the hypothalamus detect changes in blood osmolality, and sensory neurons that send action potentials through the vagus nerves to the hypothalamus detect changes in blood pressure.
2. An increase in osmolality and a decrease in blood pressure increase action potentials in ADH-secreting neurons.
3. Action potentials are carried by axons of ADH-secreting neurons through the hypothalamohypophysial tract to the posterior pituitary.
4. In the posterior pituitary, action potentials cause the release of ADH from the axon terminals into the circulatory system.
5. Increasing ADH acts on the kidney tubules to increase water reabsorption, resulting in a reduced urine volume, increased urine osmolality, and decreased blood osmolality. This helps maintain blood osmolality and volume.



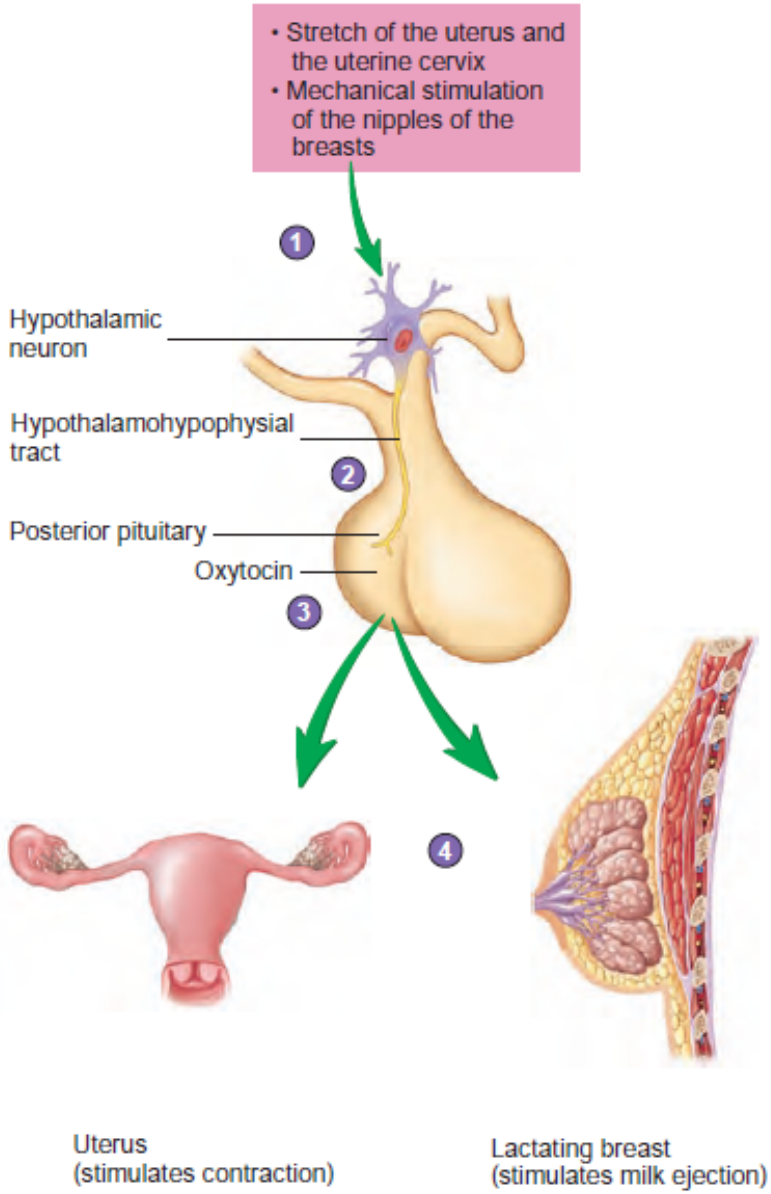
- Small changes in blood osmolality are important in regulating ADH secretion.
- Larger changes in blood pressure are required to influence ADH secretion.

Diabetes insipidus (DI) *vs.* Syndrome of inappropriate ADH secretion (SIADH)

Seeley et al., 2008

Control of Oxytocin Secretion

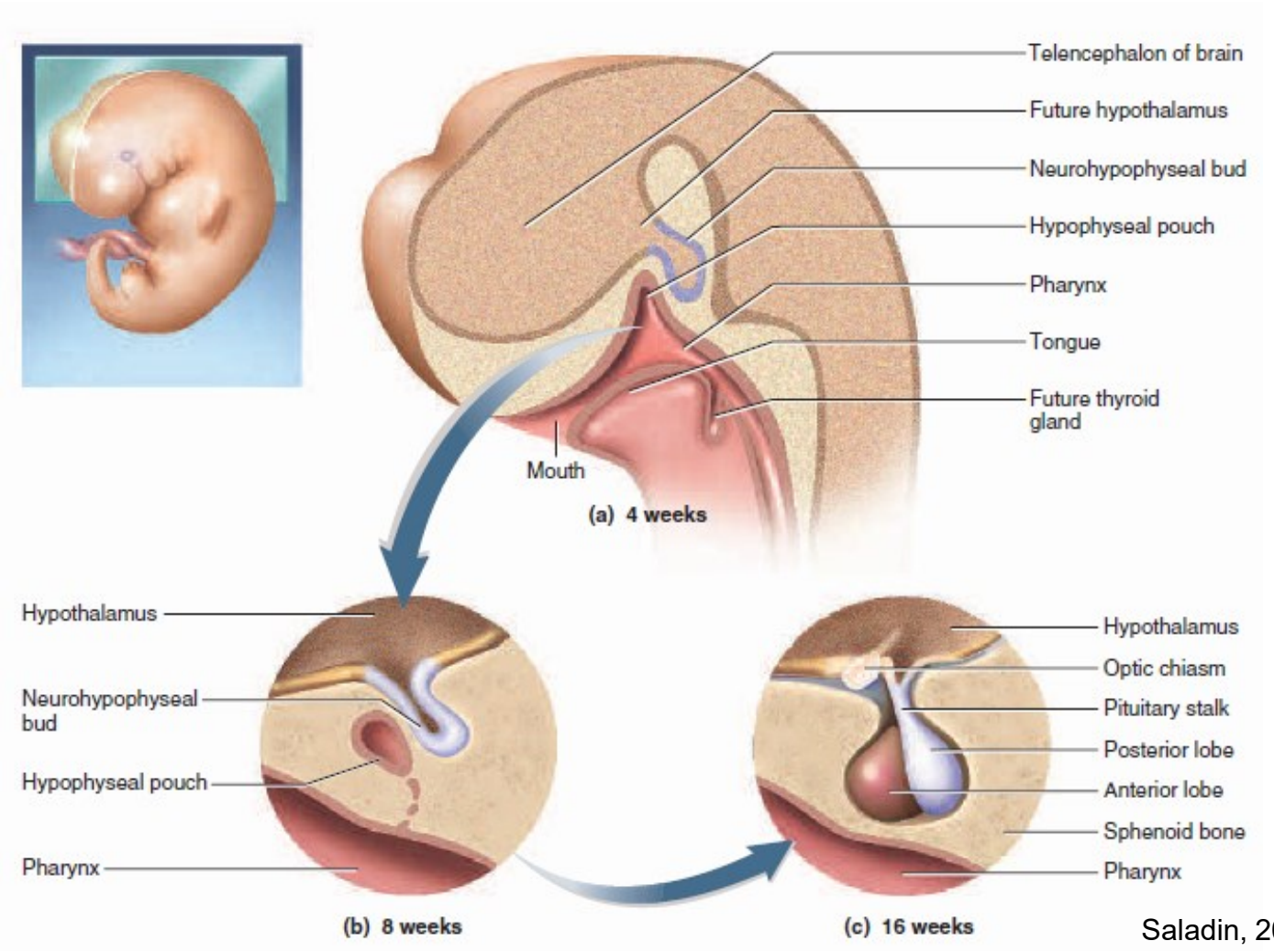
1. Action potentials are carried by sensory neurons from the uterus and breast to the spinal cord and up spinal nerve tracts to the hypothalamus.
2. Stretch of the uterus and the uterine cervix or nipples of the breasts increases action potentials in axons of oxytocin-secreting neurons.
3. Action potentials are carried by axons of oxytocin-secreting neurons in the hypothalamohypophysial tract to the posterior pituitary, where they increase oxytocin secretion.
4. Oxytocin enters the circulation. It increases contractions of the uterus and milk ejection from the lactating breast.



Seeley et al., 2008

Anterior pituitary gland

Adenohypophysis is derived from a pouch of epithelial tissue (Rathke's pouch) that migrates upward from the embryonic mouth



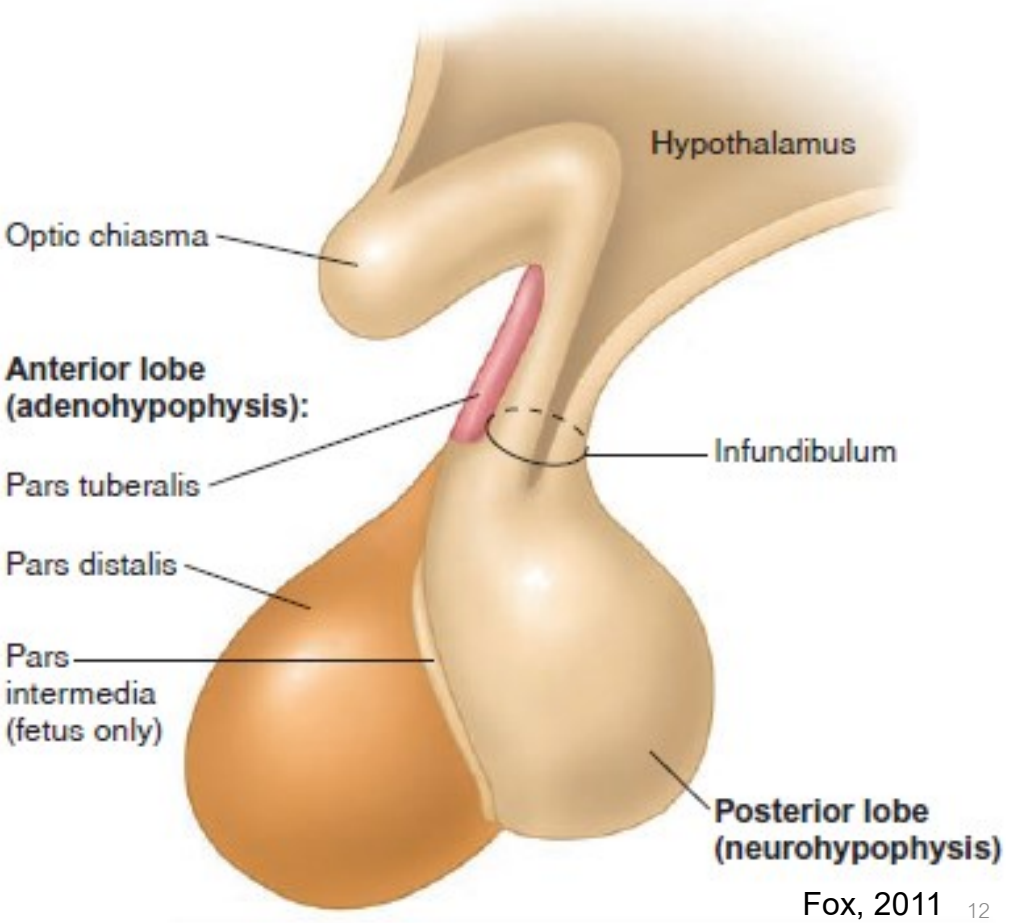
Saladin, 2009

Consists of 2 parts in adults:

(1) Pars distalis (anterior pituitary): rounded portion & major endocrine part of the gland

(2) Pars tuberalis: sheath, partially, wraps around the infundibulum

Pars intermedia: a strip between the anterior and posterior lobes (exists in the fetus)



Fox, 2011

- **Tropic hormones**
  - **adrenocorticotropic hormone (ACTH)**
  - **thyroid-stimulating hormone (TSH)**
  - **growth hormone (GH)**
  - **gonadotropins: luteinizing hormone (LH) & follicle-stimulating hormone (FSH)**
- Under hypothalamic regulation by the **hypophysiotropic neuropeptides released in the median eminence**
- Hormones produced by the target organs affect anterior pituitary function & the release of hypophysiotropic neuropeptides: **feedback control system**.

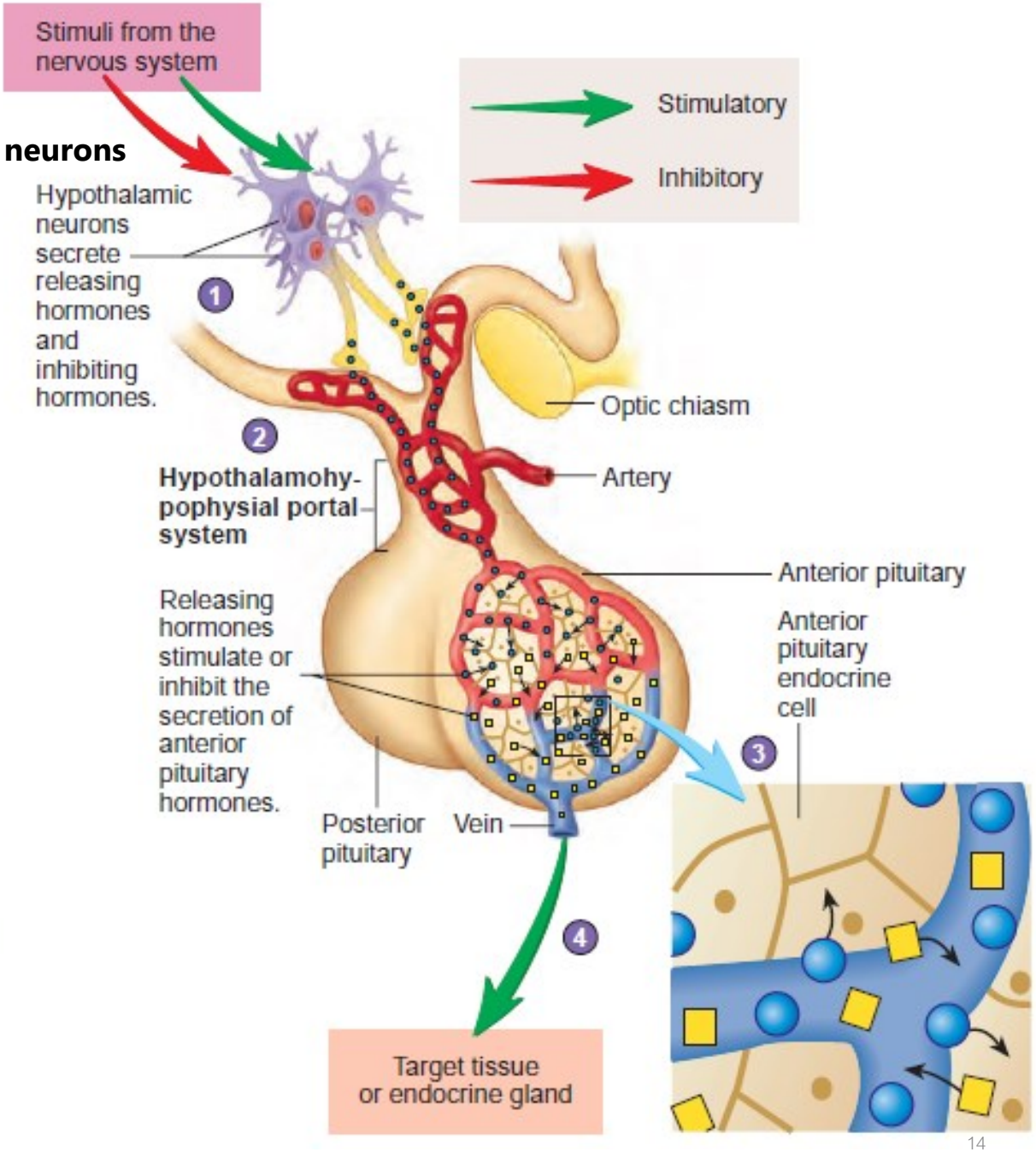
Parvocellular neurons

1. Stimuli within the nervous system increase or decrease the secretion of releasing hormones and inhibiting hormones (*blue balls*) from neurons of the hypothalamus.

2. Releasing hormones and inhibiting hormones pass through the hypothalamohypophysial portal system to the anterior pituitary.

3. Releasing hormones and inhibiting hormones leave capillaries, bind to membrane-bound receptors, and stimulate or inhibit the release of hormones (*yellow squares*) from anterior pituitary cells.

4. Anterior pituitary hormones (*yellow squares*) are carried in the blood to their target tissues (*green arrow*), which, in some cases, are other endocrine glands.



Seeley et al., 2008

Hormone-secreting cell types

Table 5.3 Hormone-secreting cell types of the anterior pituitary

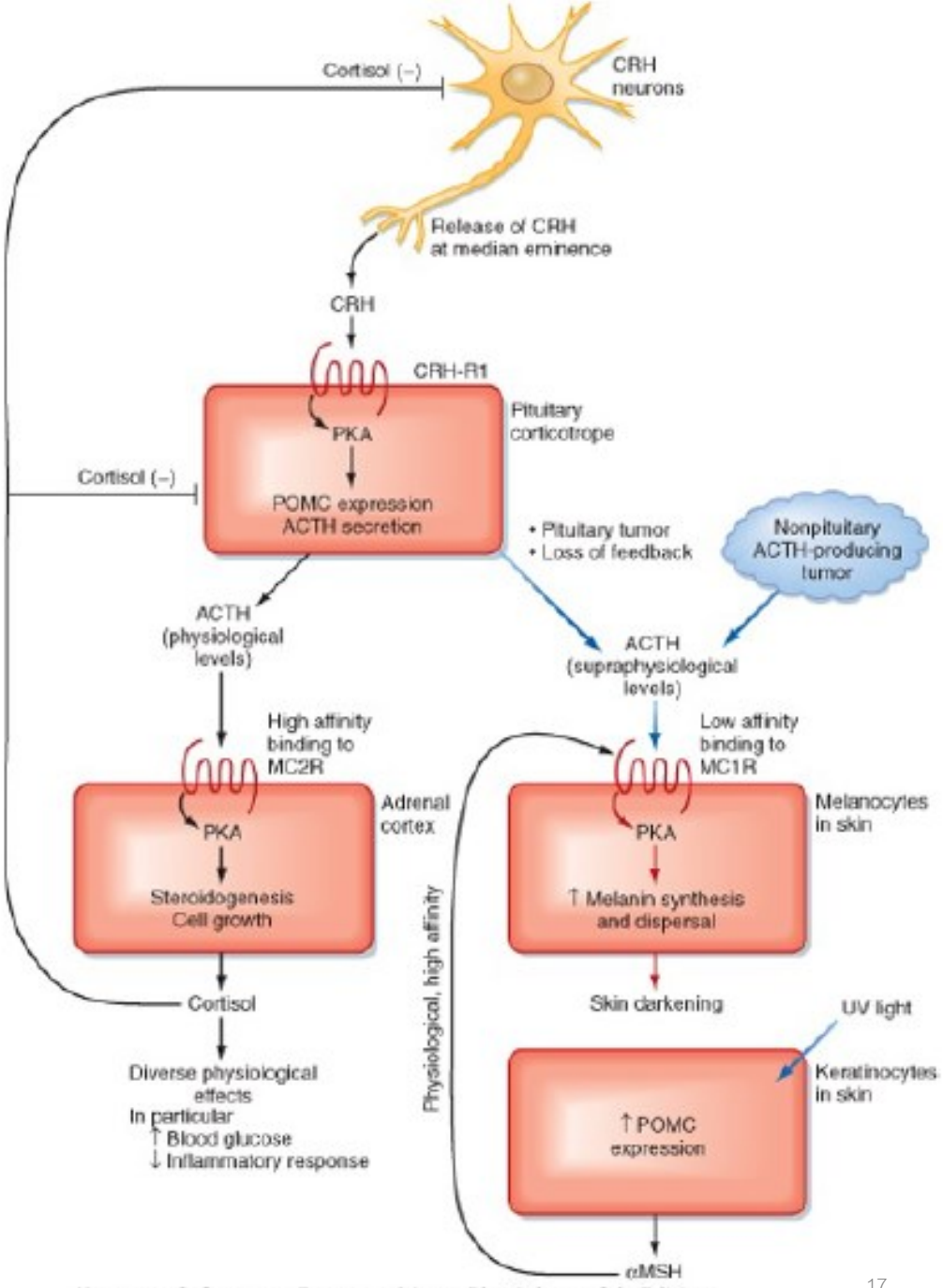
Anterior pituitary cell type	Hormone secreted	Size (number of amino acids)	Target organ	Hypothalamic regulator (+ or – effect)
Somatotroph	Growth hormone (GH)	191	Diverse	GH-releasing hormone (GHRH, +) and somatostatin (SS, –)
Lactotroph	Prolactin (PRL)	199	Breast	Dopamine (–) and thyrotrophin-releasing hormone (TRH, +)
Corticotroph	Adrenocorticotrophic hormone (ACTH)	39	Adrenal cortex	Corticotrophin-releasing hormone (CRH, +)
Thyrotroph	Thyroid-stimulating hormone (TSH)	204	Thyroid	TRH (+)
Gonadotroph	Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)	Both 204	Ovary or testis	Gonadotrophin-releasing hormone (GnRH, +)

Holt and Hanley, 2007

Corticotropes

- Produce **adrenocorticotropic hormone (ACTH; corticotropin)**
- Stimulate (are tropic to) the **adrenal cortex**, as part of the **hypothalamus-pituitary adrenal (HPA) axis**
- ACTH is a 39-amino acid peptide, synthesized as part of a larger prohormone, **pro-opiomelanocortin (POMC)**.
- **Corticotropes = POMC cells**

- CRH binds to CRH receptor (CRH-R1) on corticotropes.
- **ACTH binds to melanocortin-2 receptor (MC2R) on cell in adrenal cortex.**
- Keratinocytes express the POMC gene, but secrete alpha-MSH instead of ACTH.
- **Alpha-MSH binds to the MC1R on melanocytes.**
- At supraphysiologic levels (e.g., Addison disease with loss of negative feedback by cortisol), ACTH causes darkening of light-colored skin.



Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.
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Somatotropes

- Produce **GH (growth hormone/somatotropin)**
- **A major target: liver (where it stimulates IGF-1** (insulin-like growth factor I) **production)**
- Part of the **hypothalamus-pituitary-liver axis**
- GH also has several **direct actions at physiologic levels on nonendocrine organs** (e.g., ⇧ lipolysis, ⇧ blood glucose (⇩ glucose uptake), ⇧ protein synthesis, ⇩ insulin sensitivity).
- GH is a 191-amino acid protein, similar to PRL & human placental lactogen (hPL), and there is some overlap in activity among these hormones.

Pathologic conditions involving GH

- **Deficiencies: dwarfism**
- **Excesses: gigantism**
- **Laron dwarfs**: GH resistant because of a genetic defect in the expression of the GH receptor.
- **African pygmy**: have normal serum GH levels, but do not exhibit the normal rise in IGF that occur at puberty. However, the IGF-2 levels are normal. Unlike the Laron dwarfs, they do not totally lack the IGF response to GH
- GH deficiency in adults (after the epiphyses close): growth is not impaired, hypoglycemia
- **Acromegaly**: excessive GH secretion after the epiphyses close → appositional bone growth, but not further lengthening of long bones.
- Normal growth requires not only normal levels of GH but also normal levels of thyroid hormones, sex steroids & insulin.

Laron dwarfs and African pygmy





Gigantism & Acromegaly



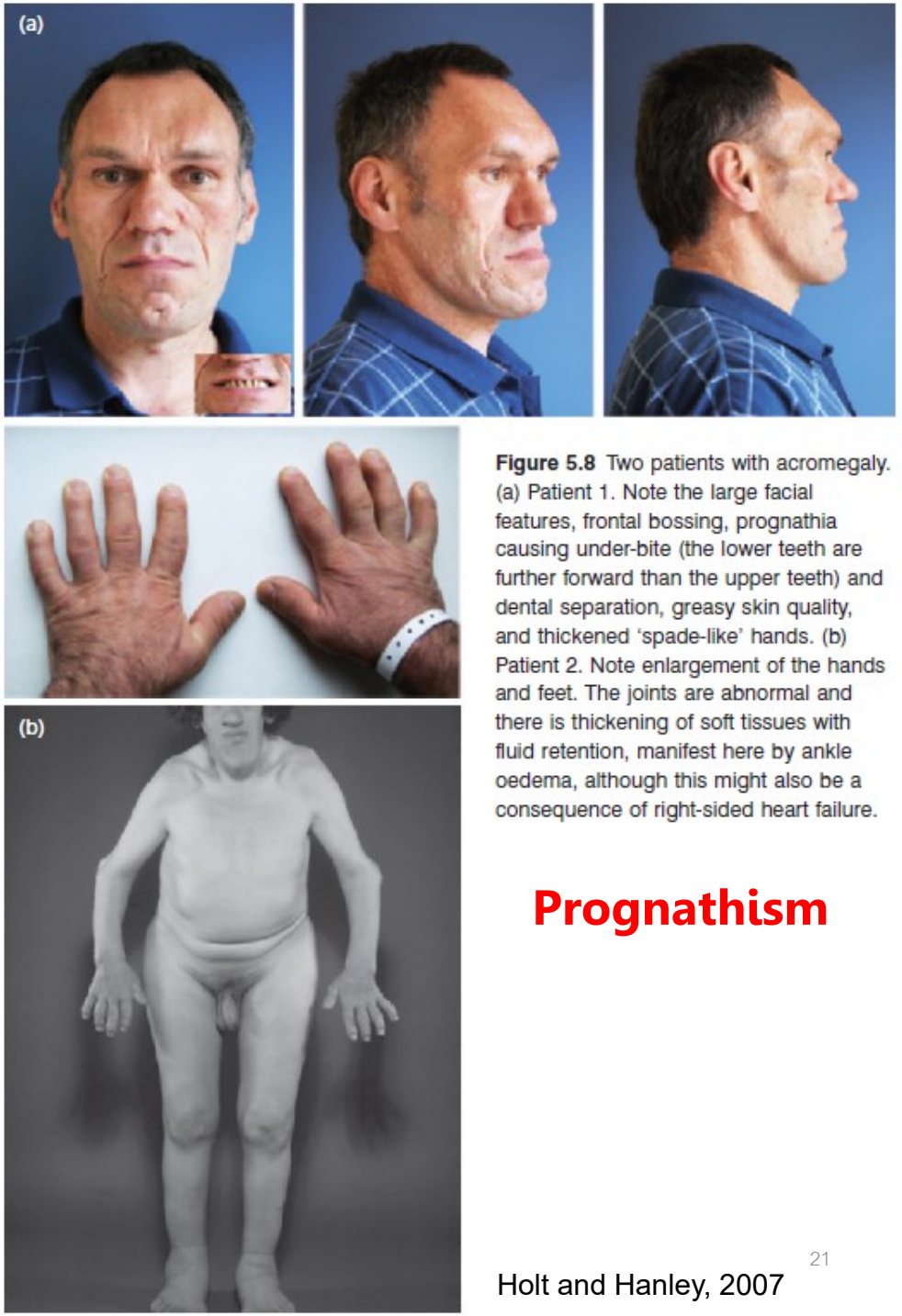


Figure 5.8 Two patients with acromegaly. (a) Patient 1. Note the large facial features, frontal bossing, prognathia causing under-bite (the lower teeth are further forward than the upper teeth) and dental separation, greasy skin quality, and thickened 'spade-like' hands. (b) Patient 2. Note enlargement of the hands and feet. The joints are abnormal and there is thickening of soft tissues with fluid retention, manifest here by ankle oedema, although this might also be a consequence of right-sided heart failure.

Prognathism

Holt and Hanley, 2007

Lactotropes

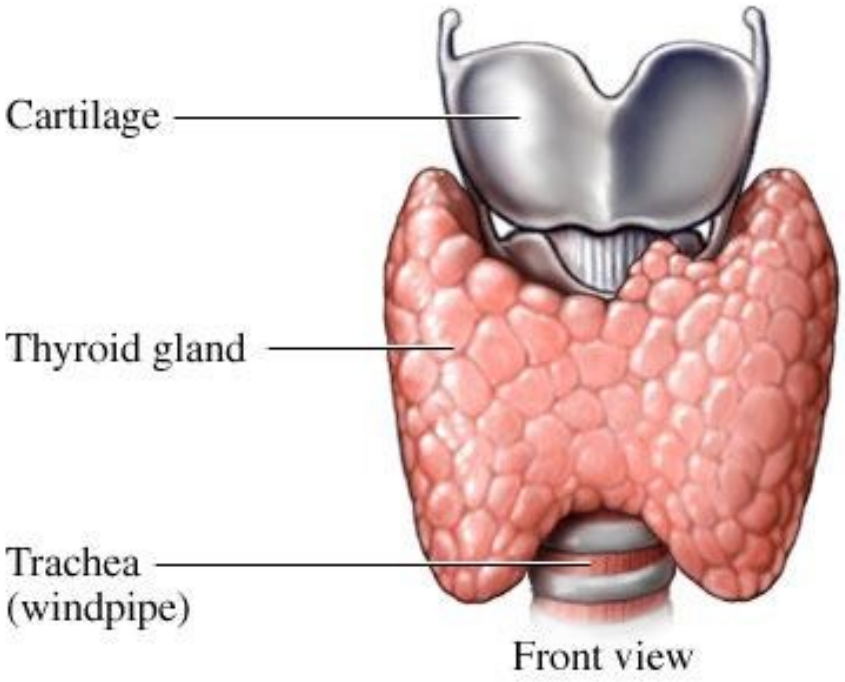
- **Lactotrope is not part of an endocrine axis → PRL acts directly on nonendocrine cells to induce physiologic changes.**
- **Production & secretion of PRL is predominantly under inhibitory control by the hypothalamus through the neurotransmitter, dopamine.**
- Disruption of pituitary stalk & the hypothalamohypophyseal portal vessels (e.g., due to surgery or physical trauma) → ⇧ PRL levels, but ⇩ ACTH, TSH, FSH, LH & GH.

TABLE 18.2 Hormones of the Pituitary Gland

Hormones	Structure	Target Tissue	Response
Posterior Pituitary (Neurohypophysis)			
Antidiuretic hormone (ADH)	Small peptide	Kidney	Increased water reabsorption (less water is lost in the form of urine)
Oxytocin	Small peptide	Uterus; mammary glands	Increased uterine contractions; increased milk expulsion from mammary glands; unclear function in males
Anterior Pituitary (Adenohypophysis)			
Growth hormone (GH), or somatotropin	Protein	Most tissues	Increased growth in tissues; increased amino acid uptake and protein synthesis; increased breakdown of lipids and release of fatty acids from cells; increased glycogen synthesis and increased blood glucose levels; increased somatomedin production
Thyroid-stimulating hormone (TSH)	Glycoprotein	Thyroid gland	Increased thyroid hormone secretion
Adrenocorticotropic hormone (ACTH)	Peptide	Adrenal cortex	Increased glucocorticoid hormone secretion
Lipotropins	Peptides	Fat tissues	Increased fat breakdown
β endorphins	Peptides	Brain, but not all target tissues are known	Analgesia in the brain; inhibition of gonadotropin-releasing hormone secretion
Melanocyte-stimulating hormone (MSH)	Peptide	Melanocytes in the skin	Increased melanin production in melanocytes to make the skin darker in color
Luteinizing hormone (LH)	Glycoprotein	Ovaries in females; testes in males	Ovulation and progesterone production in ovaries; testosterone synthesis and support for sperm cell production in testes
Follicle-stimulating hormone (FSH)	Glycoprotein	Follicles in ovaries in females; seminiferous tubes in males	Follicle maturation and estrogen secretion in ovaries; sperm cell production in testes
Prolactin	Protein	Ovaries and mammary glands in females	Milk production in lactating women; increased response of follicle to LH and FSH; unclear function in males

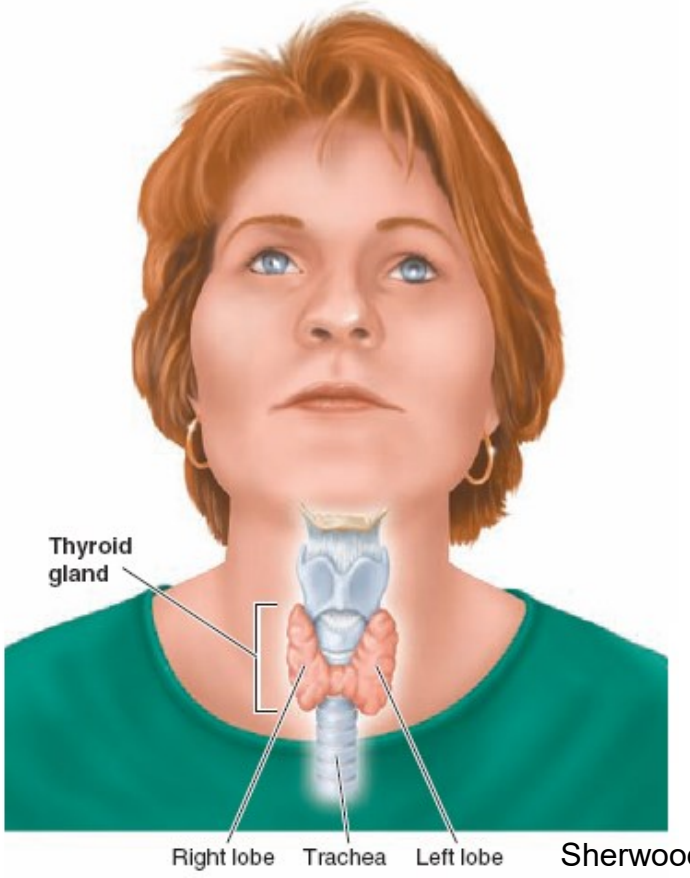
Seeley et al., 2008

Thyroid gland

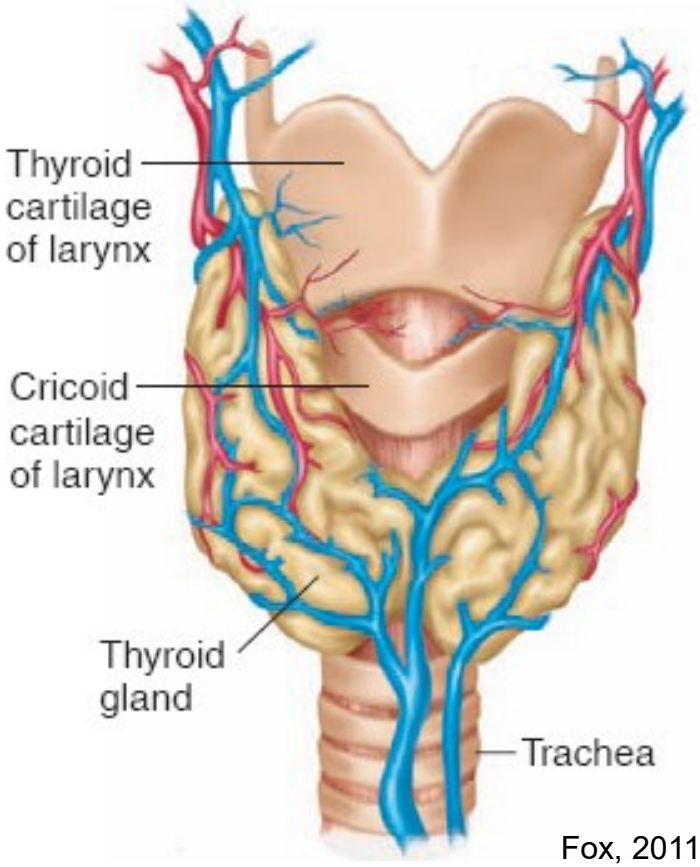


Thyroid gland

- A highly vascular, ductless alveolar/acinar/follicle gland located in the anterior neck in front of the trachea
- Consists of a right & left lobe connected by the isthmus



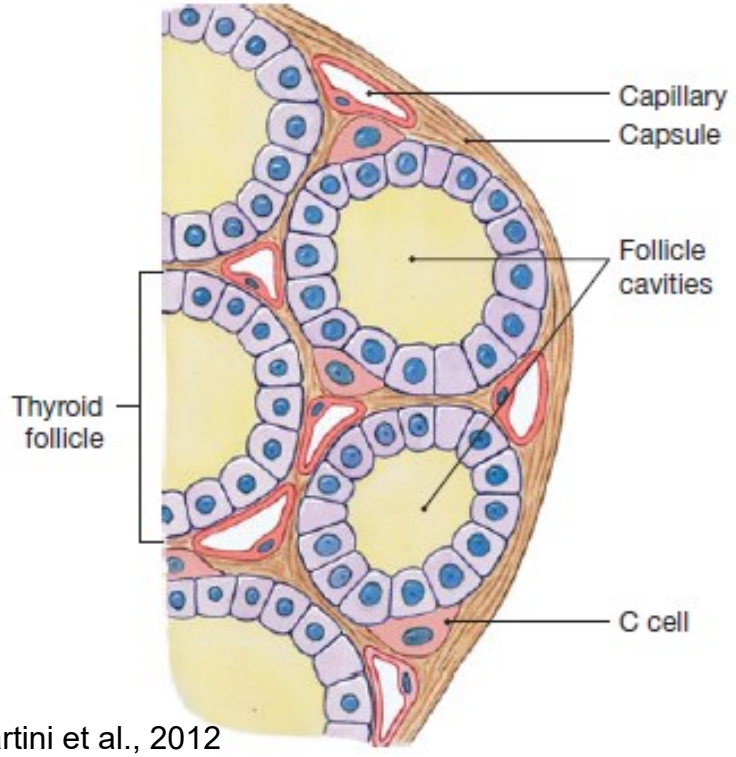
Sherwood, 2006



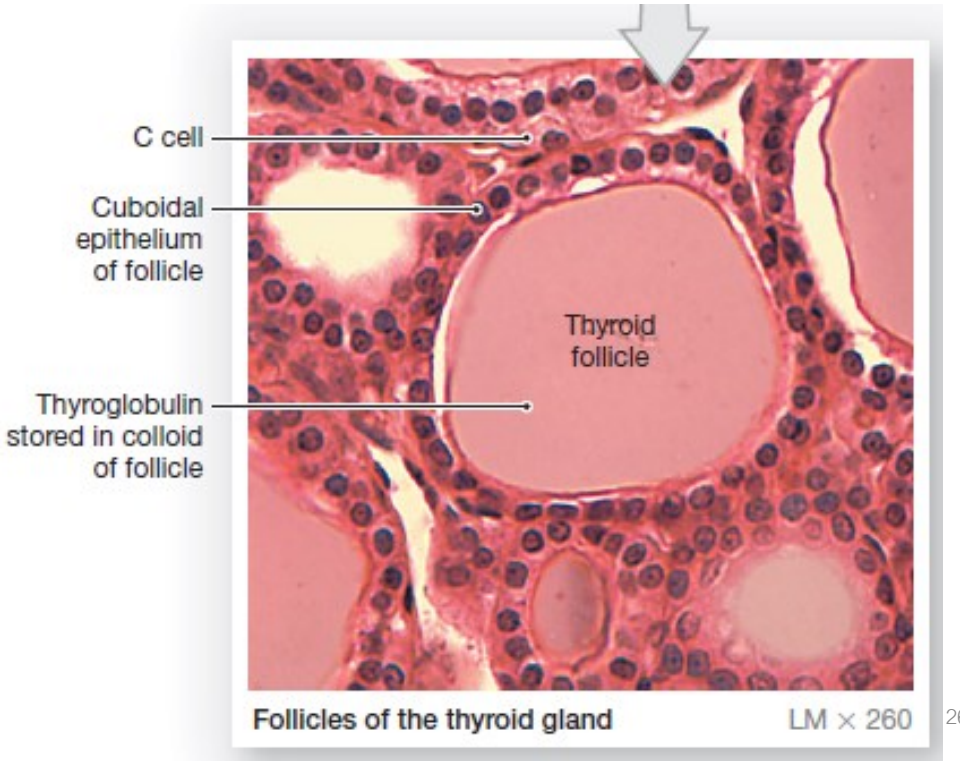
Fox, 2011

The cellular composition of thyroid gland

- **Follicular (epithelial) cells**, involved in thyroid synthesis
- **Endothelial cells** lining the capillaries, provide the blood supply to the follicles
- **Parafollicular/C cells**, involved in the production of calcitonin
- Fibroblasts, lymphocytes & adipocytes

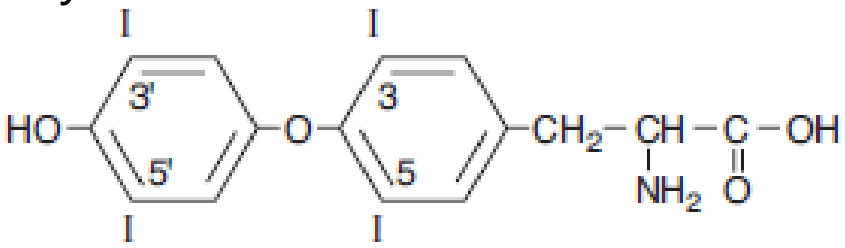


Martini et al., 2012

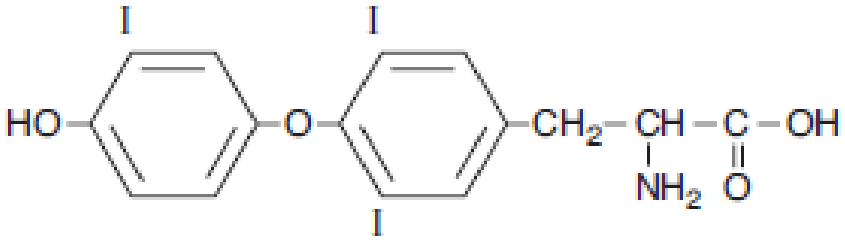


Thyroid hormone

- Primary hormone: **thyroxine (T_4)** & **triiodothyronine (T_3)** (less)
- T_3 has much greater biological activity than T_4
- T_3 is generated at its site of action in peripheral tissues by deiodination of T_4.
- Both are iodine containing amino acids.
- Small amounts of **reverse triiodothyronine** (3,3',5'-triiodothyronine, **RT_3**) and other compounds are also found in thyroid venous blood.
- RT_3 is not biologically active.



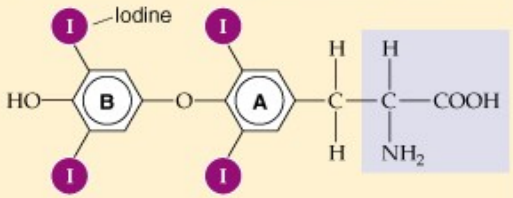
3,5,3',5',-Tetraiodothyronine (thyroxine, T_4)



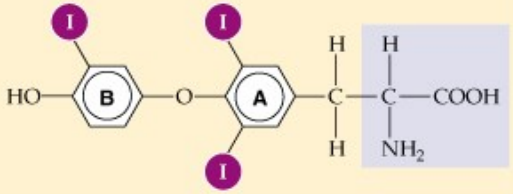
3,5,3',-Triiodothyronine (T_3)

Barrett et al., 2010

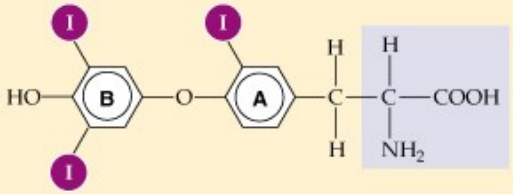
Thyroxine (T_4) (3,5,3′,5′-tetraiodo-L-thyronine)



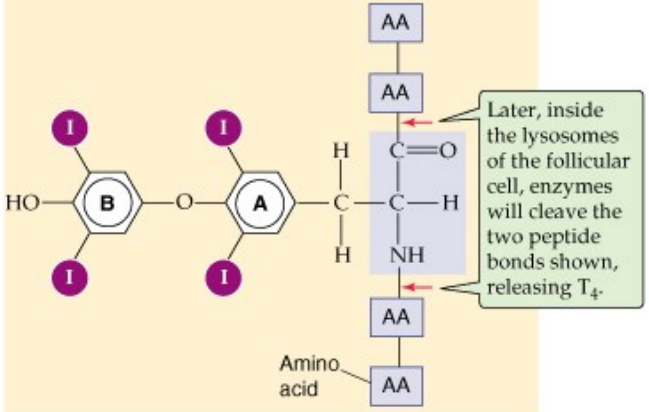
Triiodothyronine (T_3) (3,5,3′,-triiodo-L-thyronine)



Reverse T_3 (rT_3) (3,3′,5′ triiodothyronine)

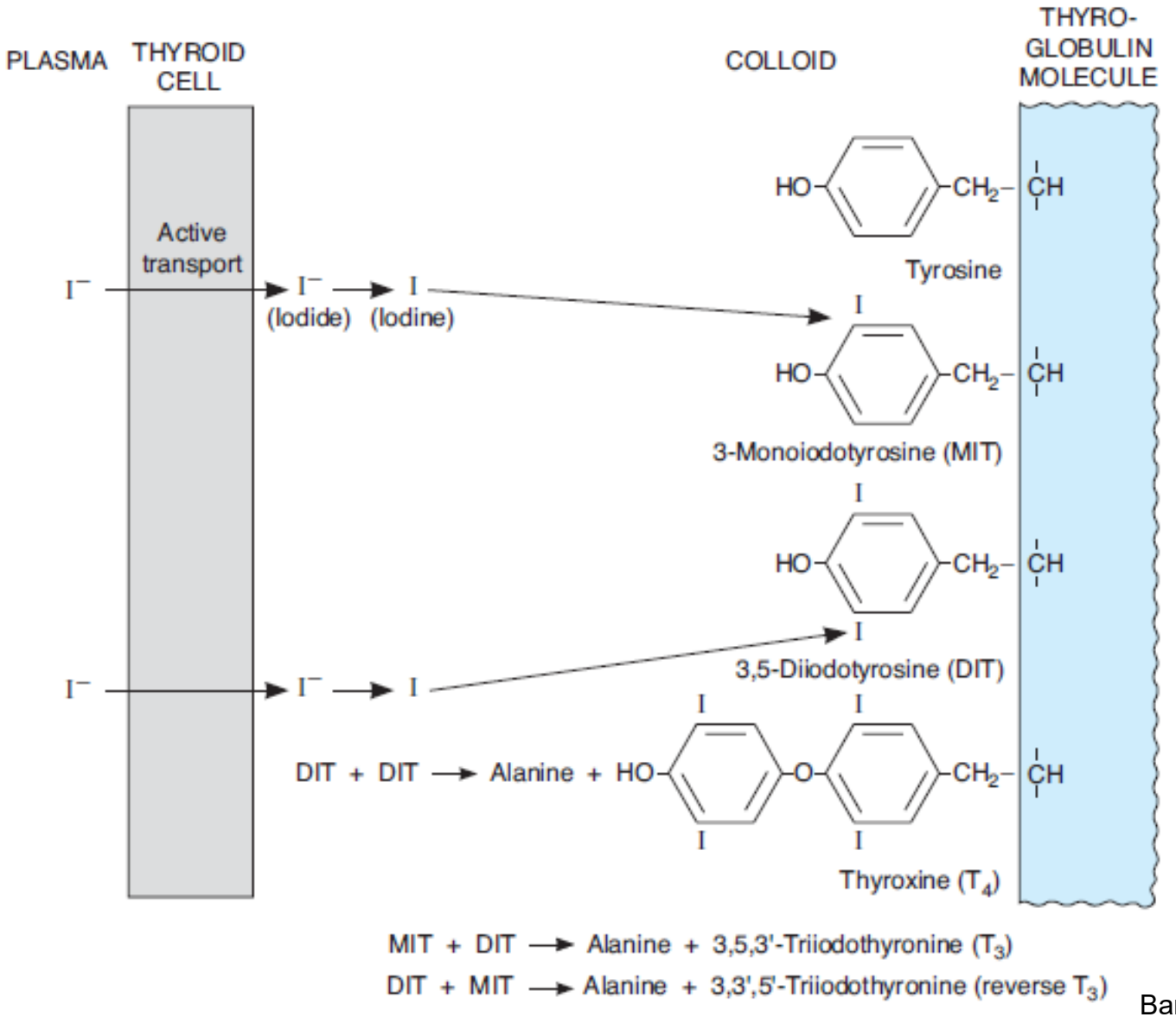


Peptide backbone of thyroglobulin molecule



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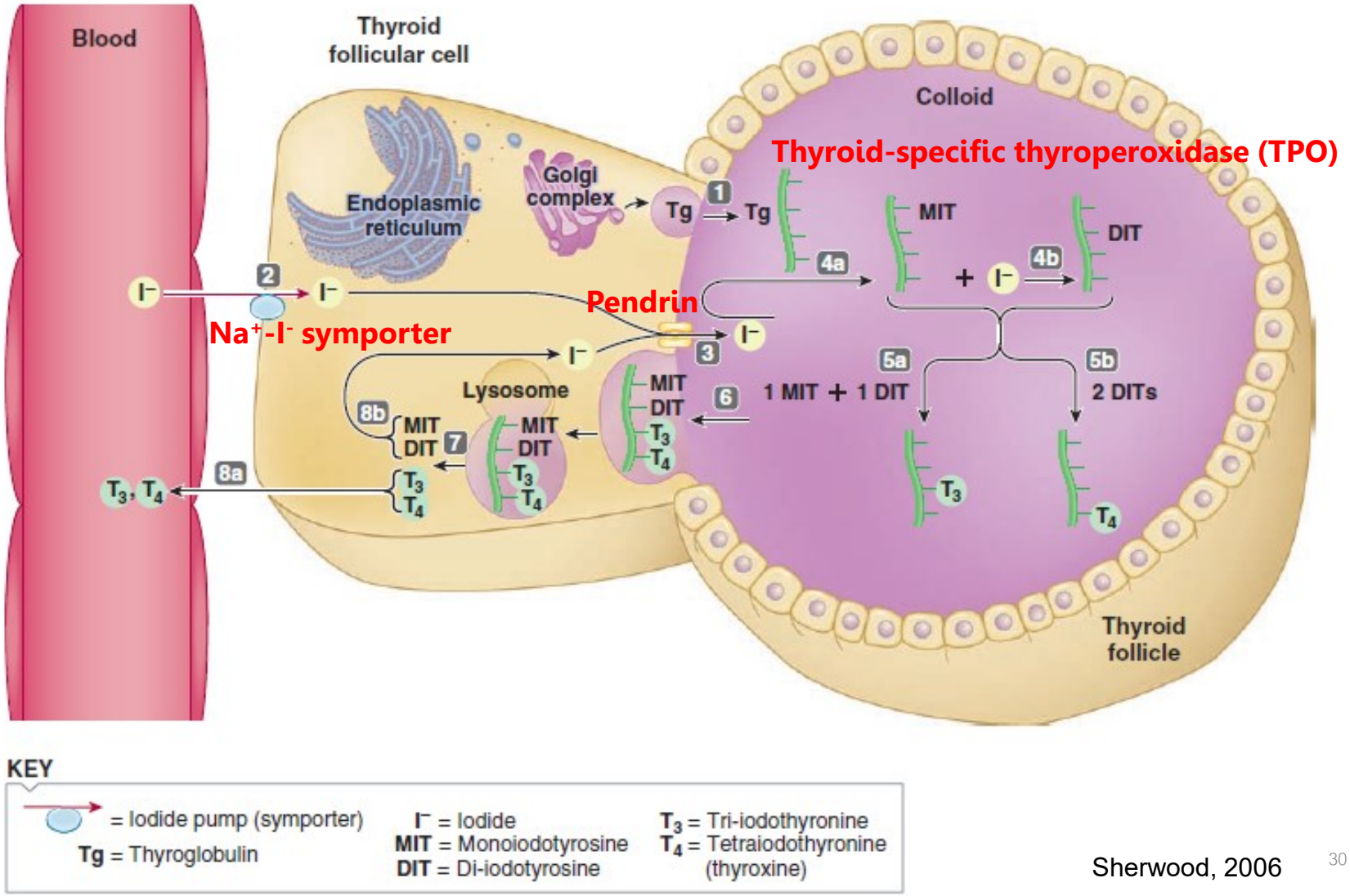
Thyroid hormone biosynthesis



- Iodine trapping
- Thyroglobulin synthesis
- Incorporation of iodine
- Coupling

Barrett et al., 2010

Synthesis, storage, and secretion of thyroid hormone



Sherwood, 2006

1. Tyrosine-containing Tg produced within the thyroid follicular cells by the ER-Golgi complex is transported by exocytosis into the colloid.

2. Iodide is carried by the energy-requiring iodide pump from the blood into the follicular cell across the basolateral membrane.

3. Iodide exits the cell through a luminal channel to enter the colloid.

4a. Attachment of 1 iodide to tyrosine within the Tg molecule yields MIT.

4b. Attachment of 2 iodides to tyrosine yields DIT.

5a. Coupling of 1 MIT and 1 DIT yields T3.

5b. Coupling of 2 DITs yields T4.

6. On appropriate stimulation, the thyroid follicular cells engulf a portion of Tg-containing colloid by phagocytosis.

7. Lysosomes attack the engulfed vesicle and split the iodinated products from Tg.

8a. T3 & T4 diffuse into the blood (secretion).

8b. MIT and DIT are deiodinated, and the freed iodide is recycled for synthesizing more hormone.

Transport of thyroid hormones

- T_4 & T_3 are lipophilic, a much larger pool of protein-bound thyroid hormones in plasma and in tissues.
- Free thyroid hormones are physiologically active and that feed back to inhibit pituitary secretion of TSH.
- The plasma proteins: **albumin, transthyretin (thyroxine-binding prealbumin) & thyroxine-binding globulin (TBG)**
- The lesser binding of T_3 correlates with T_3 has a shorter half life than T_4 and its action on the tissues is much more rapid.
- RT_3 also binds to TBG.

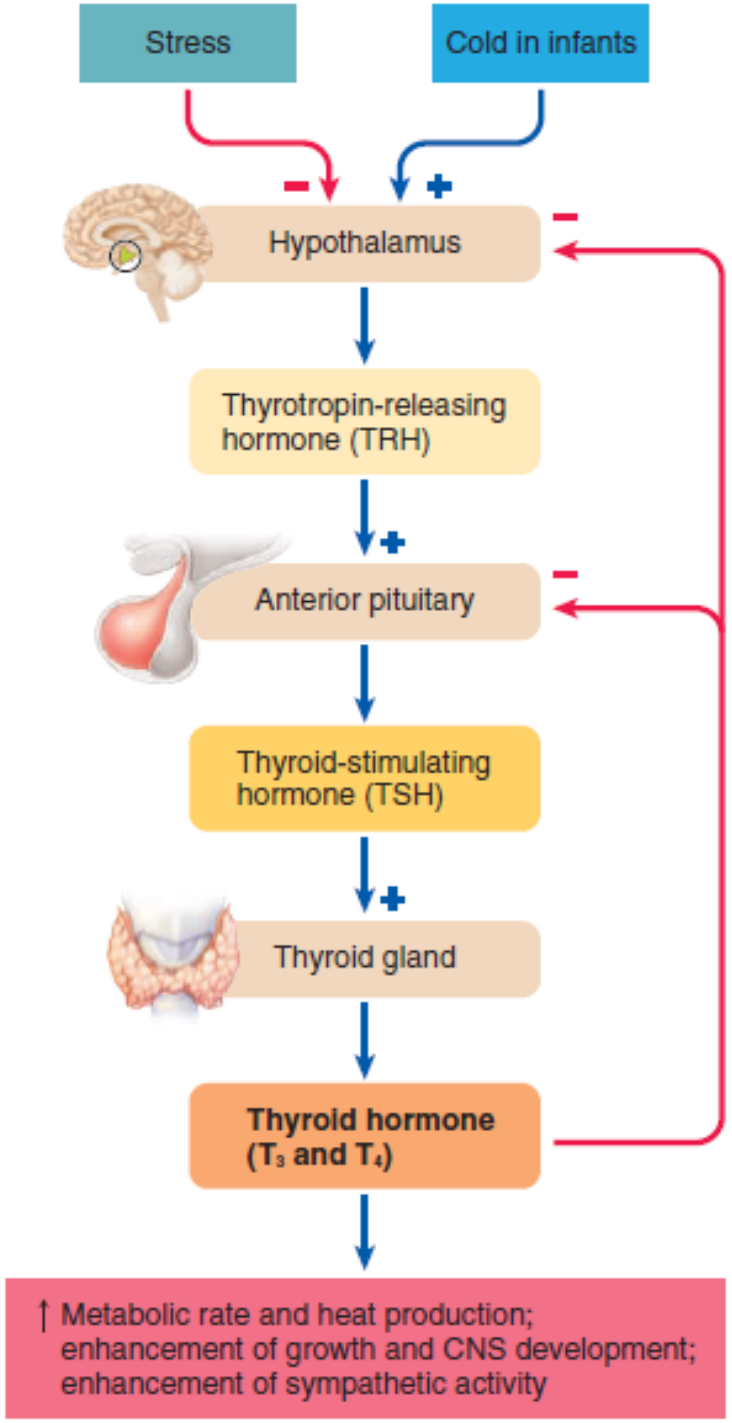
TABLE 20–1 Binding of thyroid hormones to plasma proteins in normal adult humans.

Protein	Plasma Concentration (mg/dL)	Amount of Circulating Hormone Bound (%)	
		T_4	T_3
Thyroxine-binding globulin (TBG)	2	67	46
Transthyretin (thyroxine-binding prealbumin, TBPA)	15	20	1
Albumin	3500	13	53

Barrett et al., 2010

Regulation of thyroid hormone secretion

Thyroid hormone is regulated by the **hypothalamus–pituitary–thyroid axis**



Sherwood, 2006

Effects of TSH on the thyroid

- Pituitary is removed → ⇩ thyroid function & the gland atrophies
- TSH is administered → ⇧ thyroid function
- A few minutes after the injection of TSH → ⇧ iodide binding; synthesis of T_3, T_4 & iodotyrosines; secretion of thyroglobulin into the colloid; endocytosis of colloid.
- A few hours → ⇧ Iodide trapping; blood flow
- Chronic TSH treatment → cells hypertrophy & ⇧ weight of the gland
- Enlargement of the thyroid: a **goiter**

Effects of thyroid hormone

1. Effect on metabolic rate & heat production

- ⇧ the body's overall basal metabolic rate (BMR) → ⇧ heat production

2. Sympathomimetic effect

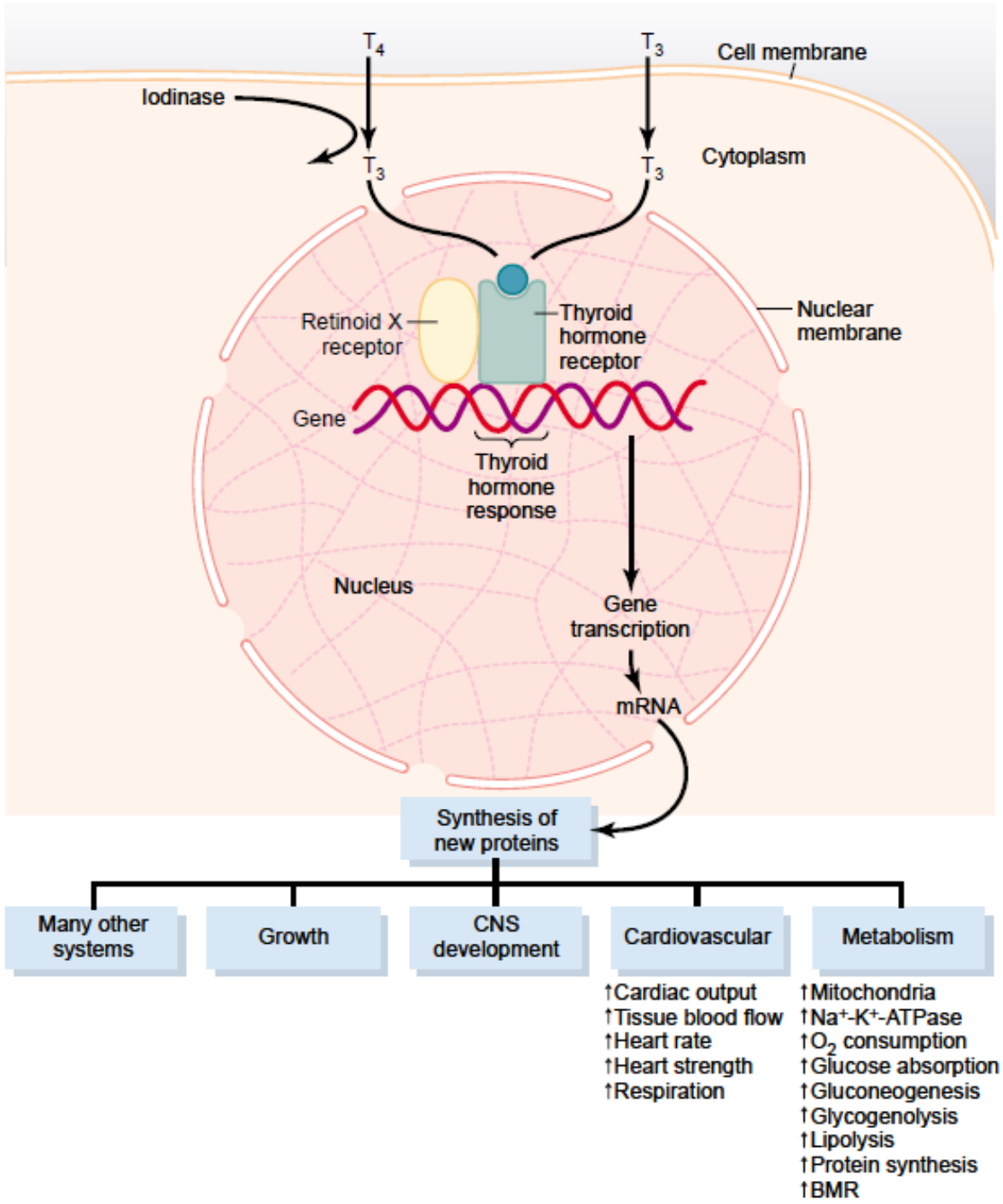
- ⇧ catecholamine target-cell receptors

3. Effect on the cardiovascular system

- ⇧ the heart's responsiveness to catecholamines

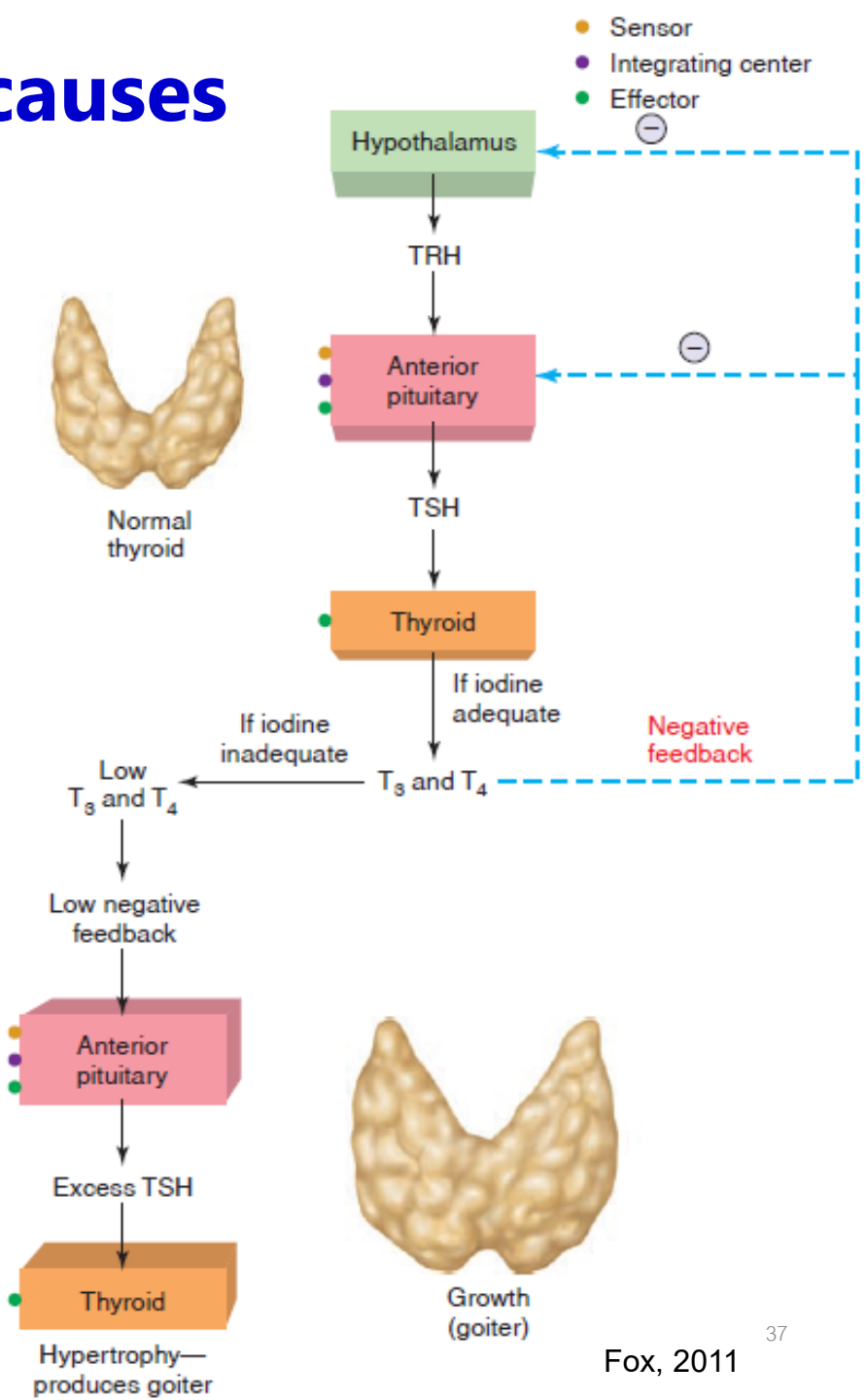
4. Effect on growth & the nervous system

- ⇧ GH secretion & IGF production by the liver
- ⇧ GH & IGF effects on new structural proteins synthesis & skeletal growth
- Thyroid-deficient children: stunted growth, reversed by thyroid replacement therapy
- Unlike excess GH, excess thyroid hormone does not produce excessive growth.
- Plays a role in the normal development of the nervous system
- Essential for normal CNS activity in adults


T4
T3
Cell membrane
Iodinase
Cytoplasm
T3
T3
Retinoid X receptor
Thyroid hormone receptor
Nuclear membrane
Gene
Thyroid hormone response
Nucleus
Gene transcription
mRNA
Synthesis of new proteins
Many other systems
Growth
CNS development
Cardiovascular
Metabolism
↑Cardiac output
↑Tissue blood flow
↑Heart rate
↑Heart strength
↑Respiration
↑Mitochondria
↑Na+-K+-ATPase
↑O2 consumption
↑Glucose absorption
↑Gluconeogenesis
↑Glycogenolysis
↑Lipolysis
↑Protein synthesis
↑BMR

How iodine deficiency causes a goiter.

Lack of adequate iodine in the diet → interferes with the negative feedback control of TSH secretion → formation of an endemic goiter



Fox, 2011

Hypothyroidism

(1) primary failure of the thyroid gland

Ex. Hashimoto disease (autoimmune disease; immune system cells attack the thyroid gland)

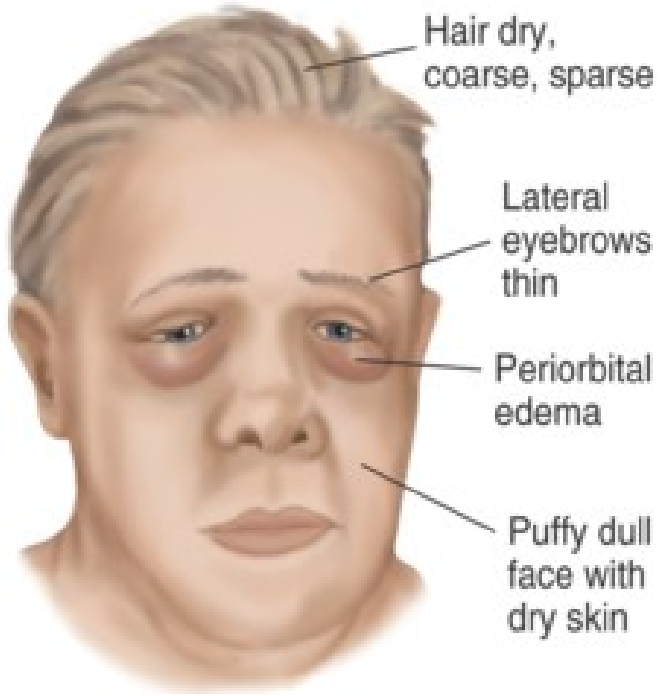
(2) secondary to a deficiency of TRH, TSH, or both

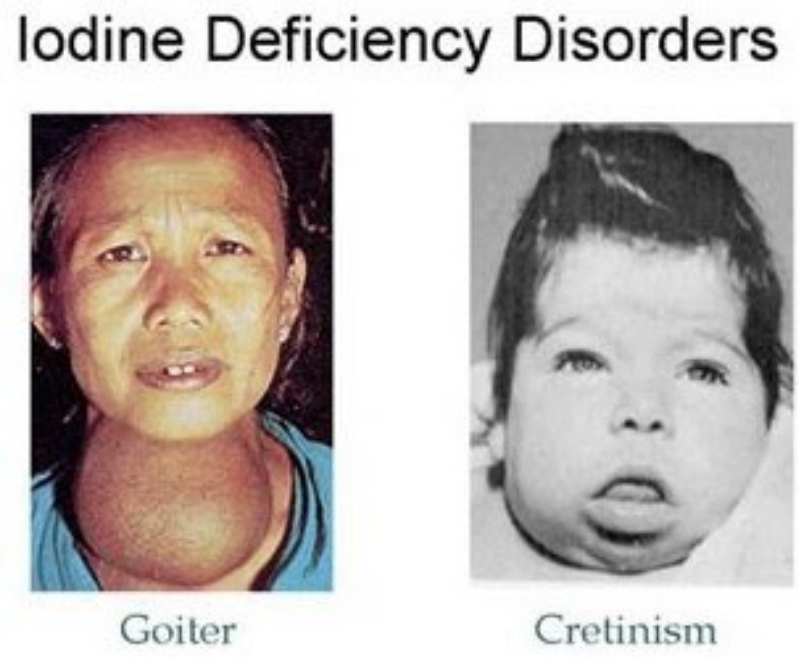
(3) an inadequate dietary supply of iodine

Symptoms:

- ⇩ BMR (less energy expenditure at rest)
- poor tolerance of cold (lack of the calorigenic effect)
- a tendency to gain excessive weight
- easily fatigued (lower energy production)
- a slow, weak pulse (⇩ HR & strength of cardiac contraction → ⇩ CO)
- slow reflexes & mental responsiveness (the effect on the nervous system)
- mental effects: ⇩ alertness, slow speech & poor memory

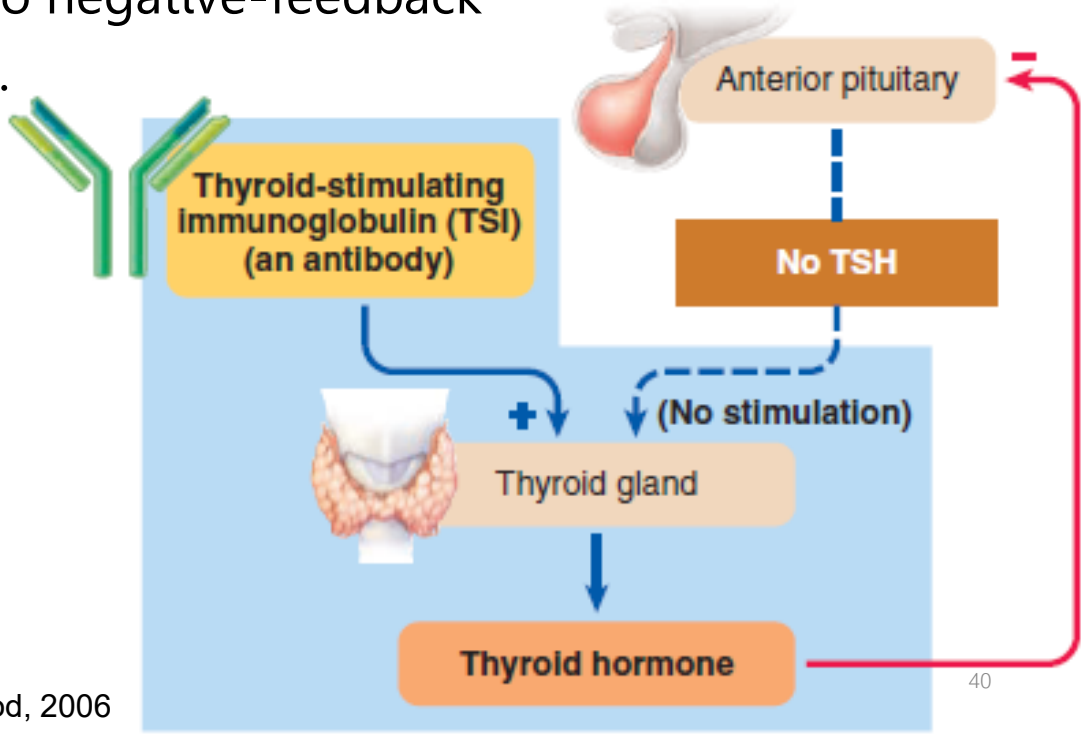
- puffy appearance/an edematous condition (primarily of the face, hands, and feet) caused by infiltration of the skin with complex, water-retaining carbohydrate molecules (a result of altered metabolism): **myxedema**
- If a person has hypothyroidism from birth, **cretinism** develops which is characterized by dwarfism, mental retardation, and general symptoms of thyroid deficiency.





Hyperthyroidism

- The most common cause: **Graves' disease**
- An autoimmune disease, the body erroneously produces **thyroid-stimulating immunoglobulin (TSI)/long-acting thyroid stimulator (LATS),** an antibody whose target is the TSH receptors on the thyroid cells.
- TSI → ⇧ secretion & growth of the thyroid (similar to TSH)
- Unlike TSH, TSI is not subject to negative-feedback inhibition by thyroid hormone.

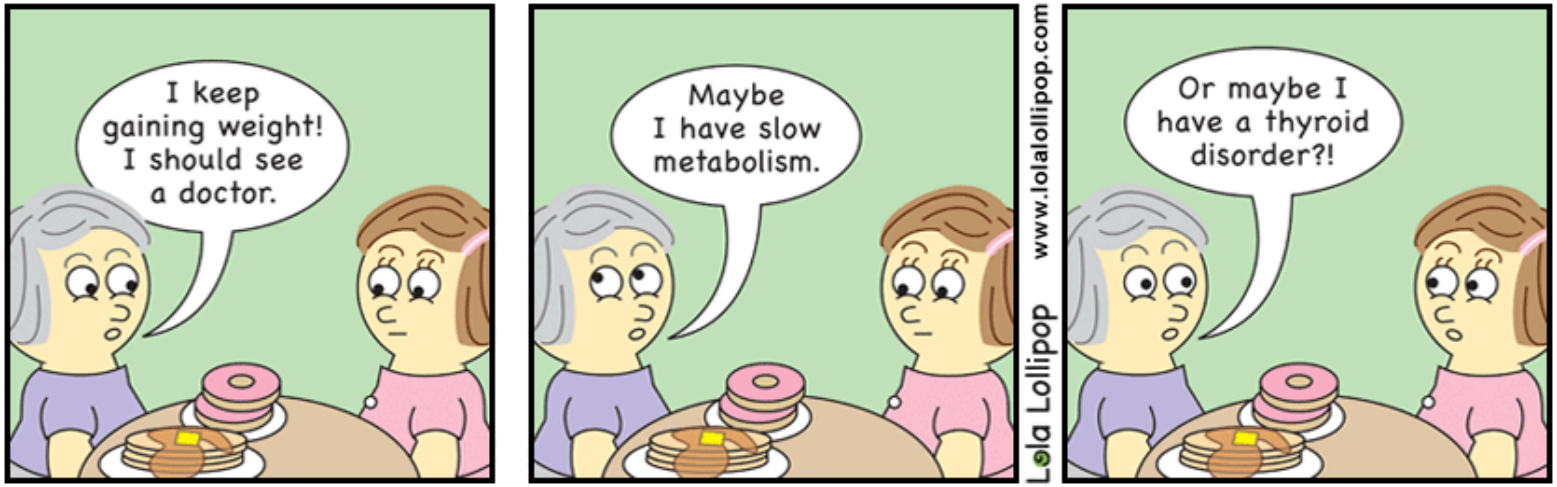


Sherwood, 2006

- Less frequently, hyperthyroidism occurs to excess TRH or TSH or in association with a hypersecreting thyroid tumor.
- The hyperthyroid patient → ⇧ BMR
- ⇧ heat production → ⇧ perspiration & poor tolerance of heat
- ⇩ body weight (the body is burning fuel at an abnormally rapid rate)
- ⇧ HR and strength of contraction → palpitations
- A prominent feature of Graves' disease, but not of the other types of hyperthyroidism, is **exophthalmos** (bulging eyes). Accumulation of hyaluronate, a glycosaminoglycan produced by fibroblast in the extraocular tissues and muscles → pushes the eyeballs forward → bulge from their bony orbit



Fox, 2011



"There is such a thing as a slow metabolism. But it's rare. The bottom line is: Most overweight or obese individuals do not get that way because of a slow metabolism. They get that way because they eat too many calories and don't exercise enough." – MAYO CLINIC